

Probability and Statistics: Problems and Solutions

Q1. Uniform Distribution on a Line Segment

Problem: A point is chosen at random on a line segment of length L . Find the probability that the ratio of the shorter to the longer segment is less than $1/4$.

Solution: Let the point be chosen at a distance x from one end, where $0 < x < L$. This forms two segments of lengths x and $L - x$. The ratio R is defined as:

$$R = \frac{\min(x, L - x)}{\max(x, L - x)}$$

We analyze two cases based on the midpoint:

- **Case 1** ($x \leq L/2$): The shorter segment is x and the longer is $L - x$.

$$\frac{x}{L - x} < \frac{1}{4} \Rightarrow 4x < L - x \Rightarrow 5x < L \Rightarrow x < \frac{L}{5}$$

- **Case 2** ($x > L/2$): The shorter segment is $L - x$ and the longer is x .

$$\frac{L - x}{x} < \frac{1}{4} \Rightarrow 4L - 4x < x \Rightarrow 5x > 4L \Rightarrow x > \frac{4L}{5}$$

Result: The valid region for x is $(0, L/5) \cup (4L/5, L)$. The total length of this region is $2L/5$.

$$P = \frac{\text{valid region}}{\text{total region}} = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Error Probability in Image Region Classification

Problem: An image has white (W) and black (B) regions. Readings from W follow $N(4, 4)$; readings from B follow $N(6, 9)$. Given a reading of 5 and a black fraction α , find α such that the probability of error is the same regardless of the conclusion.

Solution: $X|W \sim N(4, 4) \Rightarrow \sigma_1 = 2$

$X|B \sim N(6, 9) \Rightarrow \sigma_2 = 3$

$P(B) = \alpha, P(W) = 1 - \alpha$.

For equal error probability at $X = 5$, posterior probabilities must be equal:

$$P(B|X = 5) = P(W|X = 5) = \frac{1}{2}$$

Using Bayes' Theorem and the normal density formula $f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$:

$$f(5|B) = \frac{1}{3\sqrt{2\pi}}e^{-1/18}, \quad f(5|W) = \frac{1}{2\sqrt{2\pi}}e^{-1/8}$$

Substituting into the equilibrium equation:

$$\frac{\alpha \cdot f(5|B)}{\alpha \cdot f(5|B) + (1 - \alpha) \cdot f(5|W)} = \frac{1}{2}$$

$$\frac{\frac{\alpha}{3}e^{-1/18}}{\frac{\alpha}{3}e^{-1/18} + \frac{1-\alpha}{2}e^{-1/8}} = \frac{1}{2} \Rightarrow \frac{2\alpha}{3}e^{-1/18} = \frac{1-\alpha}{2}e^{-1/8}$$

$$4\alpha e^{-1/18} = 3(1-\alpha)e^{-1/8} \Rightarrow \frac{4\alpha}{3(1-\alpha)} = e^{-1/8+1/18} = e^{-5/72} \approx 0.9324$$

$$4\alpha = 2.797(1-\alpha) \Rightarrow 6.797\alpha = 2.797$$

Result: $\alpha \approx 0.38$

Q3. Minimizing Expected Distance for Facility Location

Problem: Locate a fire station at a to minimize $E[|X - a|]$.

(a) **Uniform Distribution on $(0, A)$:**

$$E[|X - a|] = \int_0^a (a - x) \frac{1}{A} dx + \int_a^A (x - a) \frac{1}{A} dx$$

Differentiating with respect to a and setting to zero:

$$\frac{d}{da} E[|X - a|] = \frac{1}{A} \left(\int_0^a 1 dx - \int_a^A 1 dx \right) = \frac{1}{A} (a - (A - a)) = \frac{2a - A}{A} = 0$$

Result: $a = A/2$ (Midpoint).

(b) **Exponential Distribution on $(0, \infty)$ with rate λ :**

$$E[|X - a|] = \int_0^a (a - x) \lambda e^{-\lambda x} dx + \int_a^\infty (x - a) \lambda e^{-\lambda x} dx$$

Differentiating: $F_X(a) - (1 - F_X(a)) = 0 \Rightarrow 2(1 - e^{-\lambda a}) - 1 = 0 \Rightarrow 1 - 2e^{-\lambda a} = 0$.

$$e^{-\lambda a} = 1/2$$

Result: $a = \frac{\ln 2}{\lambda}$ (The median).

Q4. Conditional Lifetime of Mixed Battery Types

Problem: A bin contains two battery types with exponential rates λ_1, λ_2 and probabilities p_1, p_2 . Find $P(X > t + s | X > t)$.

Solution: Using the Law of Total Probability:

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}$$

Result:

$$P(X > t + s | X > t) = \frac{p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Conditional Poisson Requests

Problem: $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ are independent. Find $P(X = k | X + Y = n)$.

Solution:

$$P(X = k | X + Y = n) = \frac{P(X = k)P(Y = n - k)}{P(X + Y = n)}$$

Substituting PMFs:

$$\frac{\left(\frac{e^{-\lambda_1} \lambda_1^k}{k!}\right) \left(\frac{e^{-\lambda_2} \lambda_2^{n-k}}{(n-k)!}\right)}{\frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^n}{n!}} = \frac{n!}{k!(n-k)!} \frac{\lambda_1^k \lambda_2^{n-k}}{(\lambda_1 + \lambda_2)^n}$$

Result: $\binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$ (Binomial distribution).

Q6. Transformation of Uniform Random Variables

Problem: $X \sim \text{Uniform}[-2, 2]$ and $Y = g(X)$, where $g(x) = x$ for $|x| \geq 1$ and $g(x) = 0$ for $|x| < 1$.

Solution (CDF $F_Y(y)$):

- $y < -2$: $F_Y(y) = 0$
- $-2 \leq y < -1$: $F_Y(y) = P(X \leq y) = \frac{y - (-2)}{2 - (-2)} = \frac{y+2}{4}$
- $-1 \leq y < 0$: $F_Y(y) = P(X \leq -1) = 1/4$
- $y = 0$: Point mass $P(Y = 0) = P(-1 < X < 1) = 2/4 = 1/2$. Thus $F_Y(0) = 1/4 + 1/2 = 3/4$.
- $0 < y < 1$: $F_Y(y) = 3/4$
- $1 \leq y \leq 2$: $F_Y(y) = P(X \leq -1) + P(-1 < X < 1) + P(1 \leq X \leq y) = \frac{1}{4} + \frac{1}{2} + \frac{y-1}{4} = \frac{y+2}{4}$

Solution (PDF $f_Y(y)$): $f_Y(y) = 1/4$ for $y \in (-2, -1) \cup (1, 2)$, with a discrete point mass $P(Y = 0) = 1/2$.

Q7. Stock Price Probability (Binomial Approximation)

Problem: Initial price S_0 . Up-move $u = 1.012$ ($p = 0.52$), down-move $d = 0.990$ ($1 - p = 0.48$). Find probability price is up $\geq 30\%$ after 1000 periods.

Solution: $S_{1000} = S_0 u^k d^{1000-k} \geq 1.30 S_0$ where k is the number of up-moves.

$$k \ln u + (1000 - k) \ln d \geq \ln(1.30)$$

$$k(\ln u - \ln d) \geq \ln(1.30) - 1000 \ln d$$

$$k \geq \frac{\ln(1.30) - 1000 \ln(0.990)}{\ln(1.012) - \ln(0.990)} \approx \frac{0.26236 - (-10.0503)}{0.011928 + 0.01005} \approx 469.4 \Rightarrow k \geq 470$$

Result: $P(k \geq 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$

Q8. Exponential Repair Times

Problem: Repair time $X \sim \text{Exp}(\lambda = 1/2)$.

(a) $P(X > 2)$: $\int_2^\infty \frac{1}{2} e^{-x/2} dx = [-e^{-x/2}]_2^\infty = e^{-1} \approx 0.3679$

(b) $P(X \geq 10 | X > 9)$: Due to memoryless property, $P(X \geq 10 | X > 9) = P(X \geq 1)$.

$$P(X \geq 1) = e^{-1/2} \approx 0.6065$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

Problem: X, Y are independent. Find PDF of Z .

Solution: $F_Z(z) = P(\sqrt{X^2 + Y^2} \leq z) = P(X^2 + Y^2 \leq z^2)$

$$F_Z(z) = \int_{-z}^z \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x)f_Y(y)dydx$$

Differentiating using Leibniz's rule and symmetry: **Result:** $f_Z(z) = z \int_{-z}^z \frac{f_X(x)[f_Y(\sqrt{z^2-x^2})+f_Y(-\sqrt{z^2-x^2})]}{\sqrt{z^2-x^2}} dx$, for $z > 0$.

Q10. Joint Distribution of Successive Maxima

Problem: $M_n = \max(X_1, \dots, X_n)$ where X_i are i.i.d. with CDF F . Find joint CDF of M_n, M_{n+1} .

Solution: Case 1 ($x \leq y$): $M_n \leq x$ and $M_{n+1} \leq y$ implies $\{X_1, \dots, X_n\} \leq x$ and $X_{n+1} \leq y$.

$$P(M_n \leq x, M_{n+1} \leq y) = F(x)^n F(y)$$

Case 2 ($x > y$): $M_{n+1} \leq y$ automatically implies $M_n \leq y$. Since $y < x$, the condition $M_n \leq x$ is redundant.

$$P(M_n \leq x, M_{n+1} \leq y) = P(M_{n+1} \leq y) = F(y)^{n+1}$$

Result:

$$P(M_n \leq x, M_{n+1} \leq y) = \begin{cases} F(x)^n F(y), & x \leq y \\ F(y)^{n+1}, & x > y \end{cases}$$